Team 2 - Annie, Emma, Lucas, Mackenzie, Haifa
Dr. Rui Yan
BTC1895H
July 10, 2026

# Patient Experience & At-Risk Analysis

*MedCore Patient Feedback Dataset*

## Executive Summary

Across 289,889 patient feedback responses collected over three years, 38.6% are flagged At-Risk. These are patients whose experiences compromise their continued relationship with MedCore. This is not a marginal or isolated issue, it is a persistent, organization-wide pattern that has remained essentially flat, between 37.9% and 39.5% every quarter, since 2023.

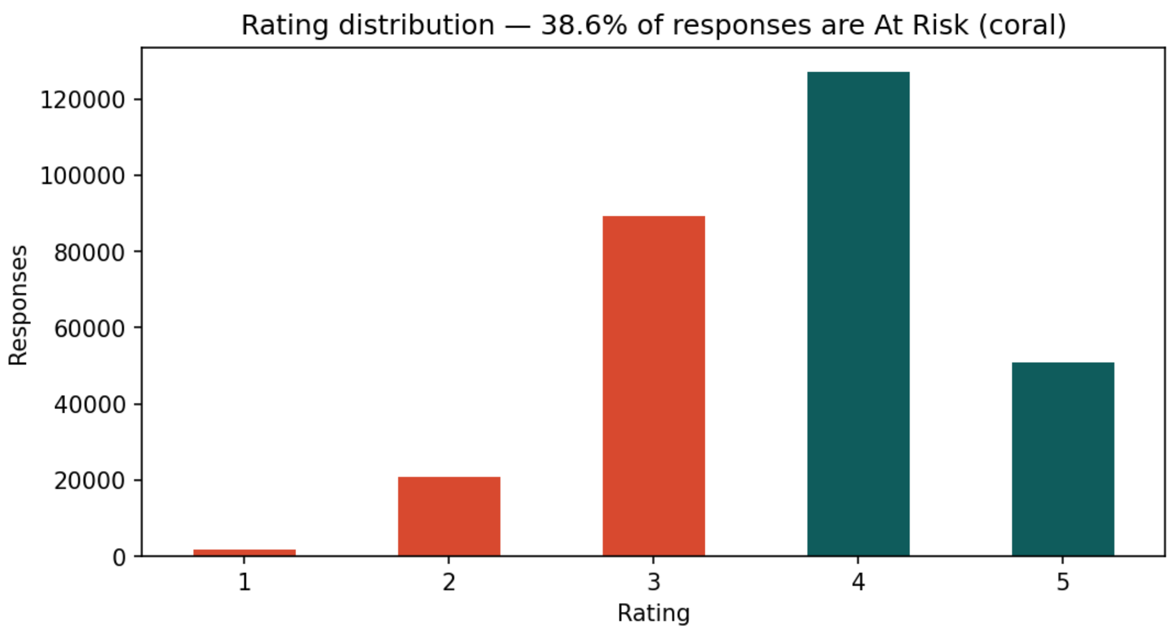
Key findings:

- Certain complaint themes are disproportionately severe. Billing, Provider, and Facilities complaints all resulted in 100% At-Risk rates, while Wait Time and Communication were the largest contributors by volume.
- The problem appears to be operational rather than structural. At-Risk rates vary substantially across departments, ranging from 20.5% to 57.9% (nearly a threefold difference), while the organization-wide rate has remained unchanged for three years. This suggests that department-level practices, rather than unavoidable differences in patient populations, are driving the results.
- Dissatisfaction has a measurable price tag. At-Risk patients no-show 6.1x more often than satisfied patients. Isolating just the no-shows attributable to dissatisfaction (above the baseline rate every patient population has) comes to an estimated $13.3M in lost visit revenue over three years, which is about $4.4M per year.

Team 2 - Annie, Emma, Lucas, Mackenzie, Haifa
Dr. Rui Yan
BTC1895H
July 10, 2026

## 1. Rating Distribution & Overall At-Risk Rate

The dataset skews positive on raw star rating (ratings of 4 and 5 make up the majority of responses) but the At-Risk flag captures a broader and more clinically meaningful signal than star rating alone. Overall, 38.6% of all 289,889 responses are flagged At-Risk, meaning more than 1 in 3 patients who give feedback are at risk of disengaging from care.

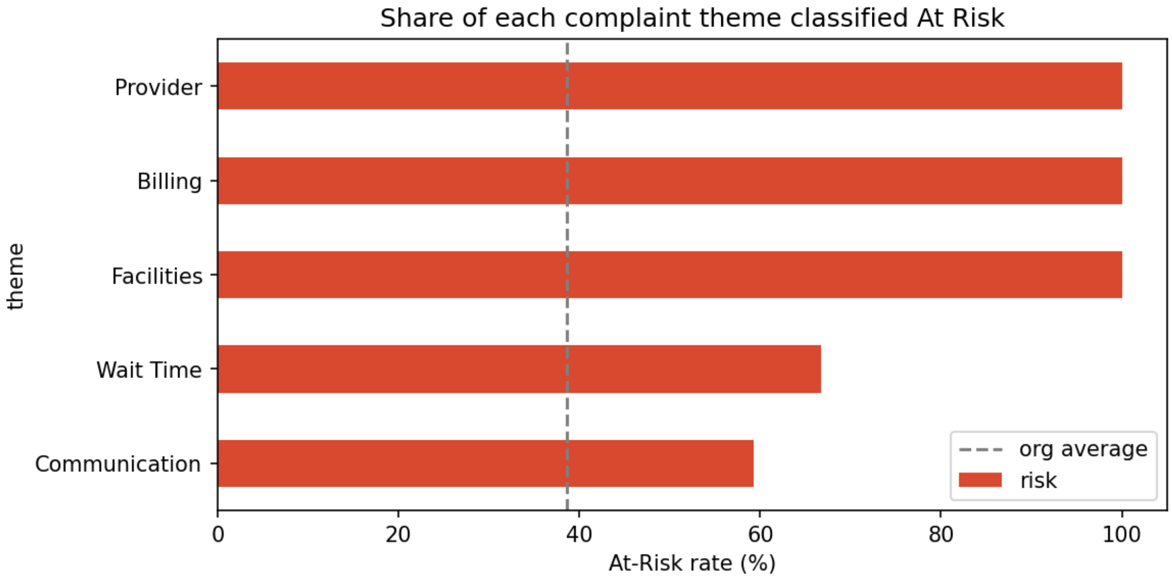



Although At-Risk is derived from patient ratings, the analyses that follow explain why patients become At-Risk by examining complaint themes, booking delays, departments, and operational factors.

## 2. At-Risk Rate by Complaint Theme

When complaints are grouped into themes, a clear severity hierarchy emerges:

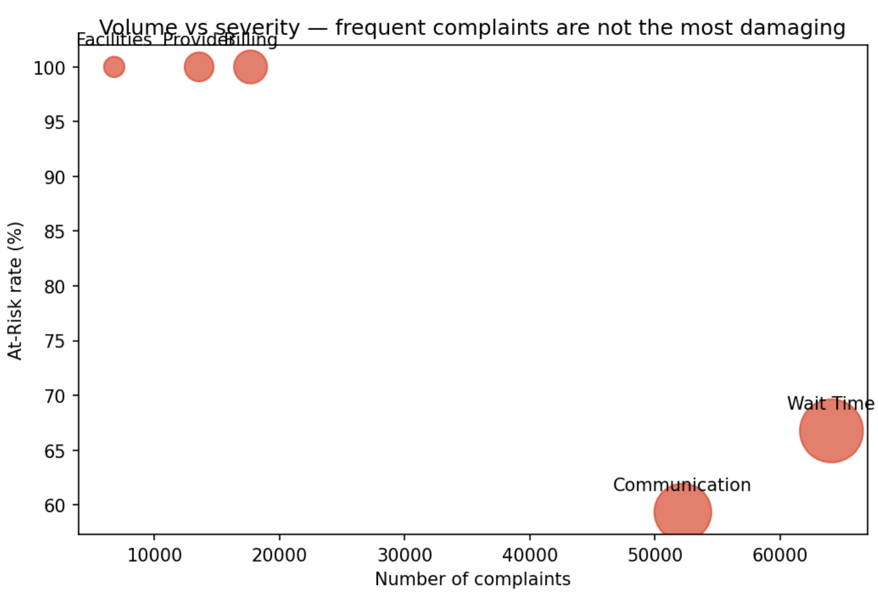
| Theme | Complaints (n) | At-Risk rate |
|---|---|---|
| Billing | 17,681 | 100.0% |
| Provider | 13,570 | 100.0% |
| Facilities | 6,785 | 100.0% |
| Wait Time | 64,108 | 66.8% |
| Communication | 52,229 | 59.3% |
| Positive / No Complaint | 135,516 | 0.0% |




Billing, Provider, and Facilities complaints are the strongest indicators of an At-Risk patient. In this dataset, every patient who reported one of these issues was classified as At-Risk. In contrast, Wait Time and Communication complaints are less likely to result in an At-Risk outcome on an individual basis, but their much higher frequency makes them a larger driver of overall dissatisfaction, as explored in the following section.

## 3. Volume vs. Severity — Where to Prioritize

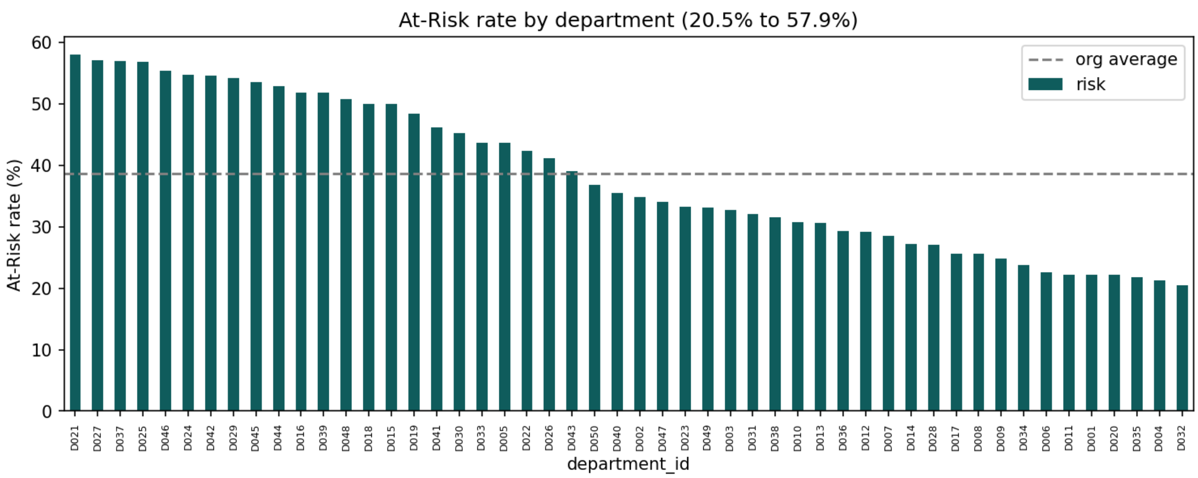
Comparing complaint volume with At-Risk conversion highlights an important trade-off: the most frequent complaints are not the most severe. Wait Time (64,108 complaints) and Communication (52,229) dominate by volume but have lower At-Risk conversion rates (67% and 59%) than Billing, Provider, and Facilities complaints, all of which have a 100% At-Risk rate. However, because Wait Time and Communication affect far more patients, improvements in these areas are likely to produce the largest overall reduction in dissatisfaction.



These findings support a two-track improvement strategy. High-volume themes such as Wait Time and Communication offer the greatest opportunity to reduce overall patient dissatisfaction because they affect the largest number of patients. In contrast, high-severity themes such as Billing, Provider, and Facilities occur less often but every reported complaint in these categories was classified as At-Risk in this dataset.

## 4. Department Hotspots

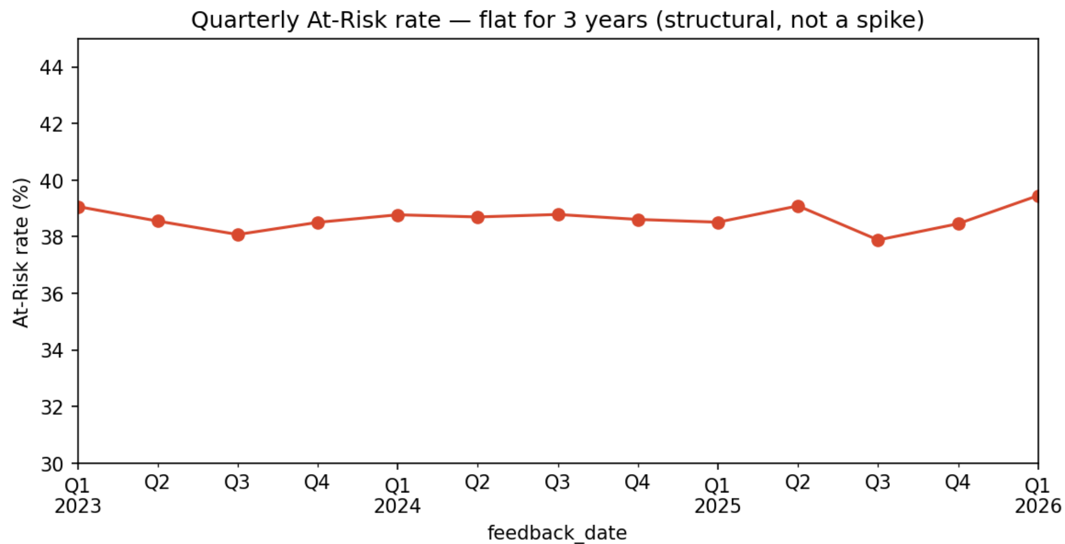
Among departments with at least 50 responses, At-Risk rates range from 20.5% to 57.9%—nearly a threefold difference between the best- and worst-performing departments. This substantial variation suggests that differences in department-level practices, such as scheduling, staffing, provider interactions, and facility management, may be contributing to patient dissatisfaction, rather than the issue being uniform across all departments.




The organization-wide average (38.6%, dashed line) masks this variation. Several departments consistently outperform the average, demonstrating that lower At-Risk rates are achievable. This indicates that targeted improvements in the highest-risk departments have the potential to meaningfully reduce overall patient dissatisfaction.

## 5. Quarterly Trend — A Structural, Not Recent, Problem

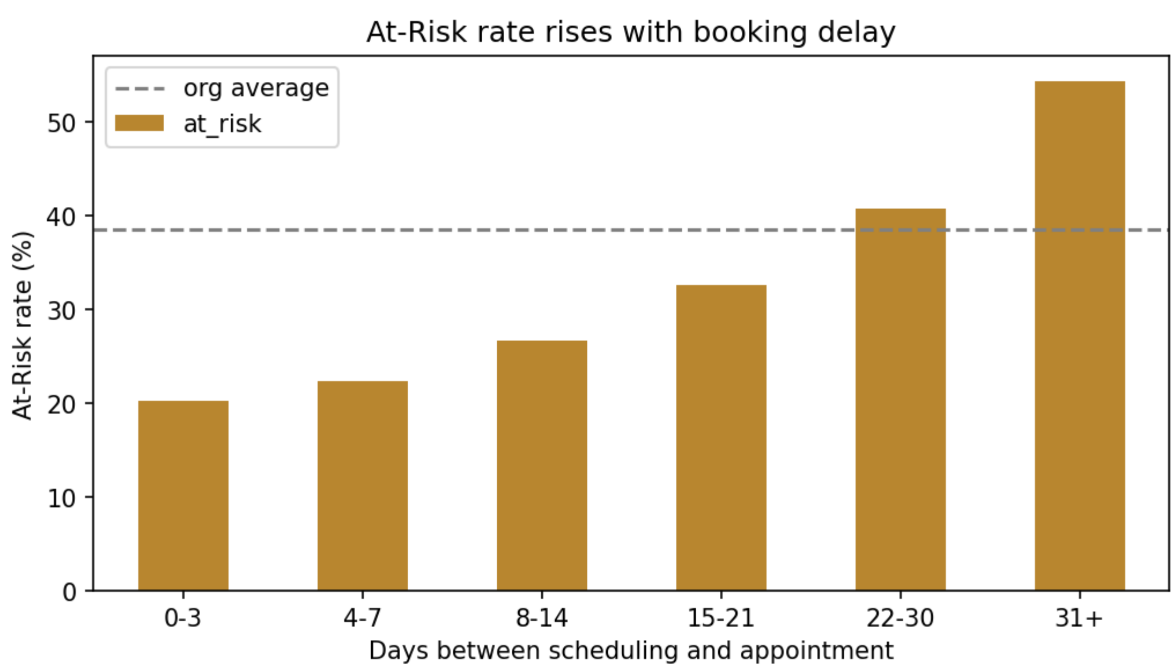
The At-Risk rate remained remarkably stable from Q1 2023 through Q1 2026, fluctuating within a narrow range of 37.9% to 39.5% with no sustained upward or downward trend.




This consistency suggests that patient dissatisfaction is a persistent organizational issue rather than the result of a temporary spike or seasonal fluctuation. The organization has maintained an At-Risk rate of approximately 39% for three years, indicating that meaningful improvement is unlikely without deliberate operational changes.

## 6. Booking Delay Is a Leading Driver

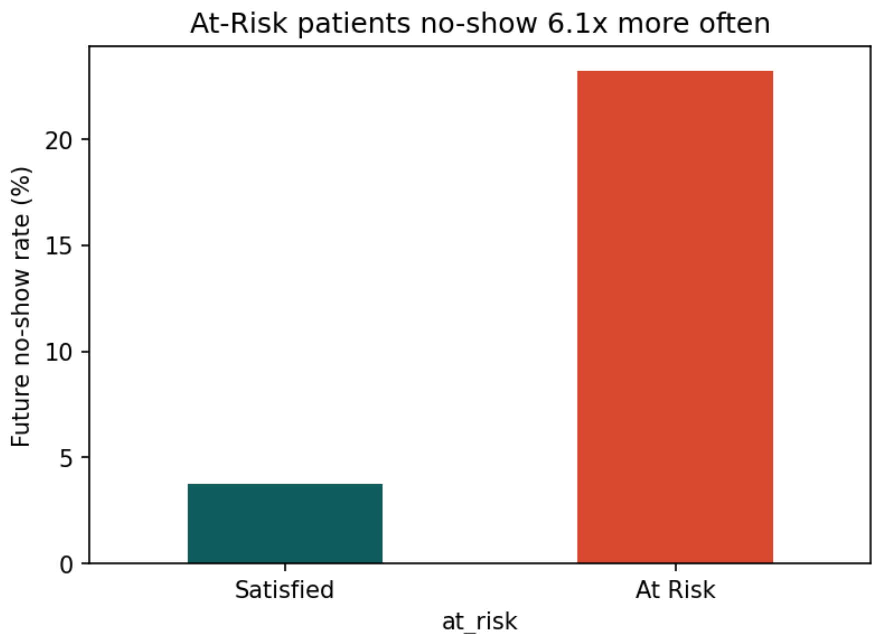
The At-Risk rate increases steadily with booking delay, rising from 20.2% for patients seen within three days to 54.3% for those waiting 31 or more days, a 2.7-fold increase across the range.



The relationship is monotonic across every waiting-time bucket, making booking delay one of the strongest operational factors associated with patient dissatisfaction in this dataset. This pattern suggests that reducing scheduling delays, even incrementally, could meaningfully lower the proportion of patients classified as At-Risk.

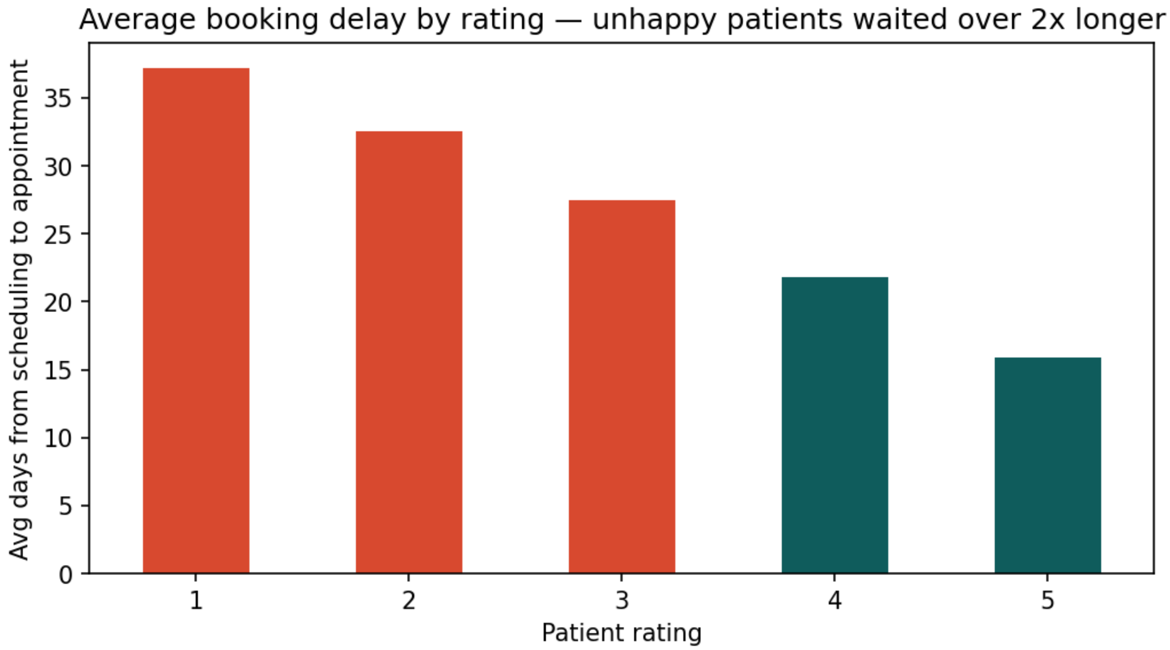
## 7. At-Risk Patients Are Far More Likely to No-Show

At-Risk patients no-show future appointments at a rate of 23.2%, compared with 3.8% for satisfied patients—a 6.1-fold difference. This finding connects patient experience to operational and financial outcomes. Patients who report dissatisfaction are substantially more likely to miss future appointments, contributing to lost visit volume and revenue.



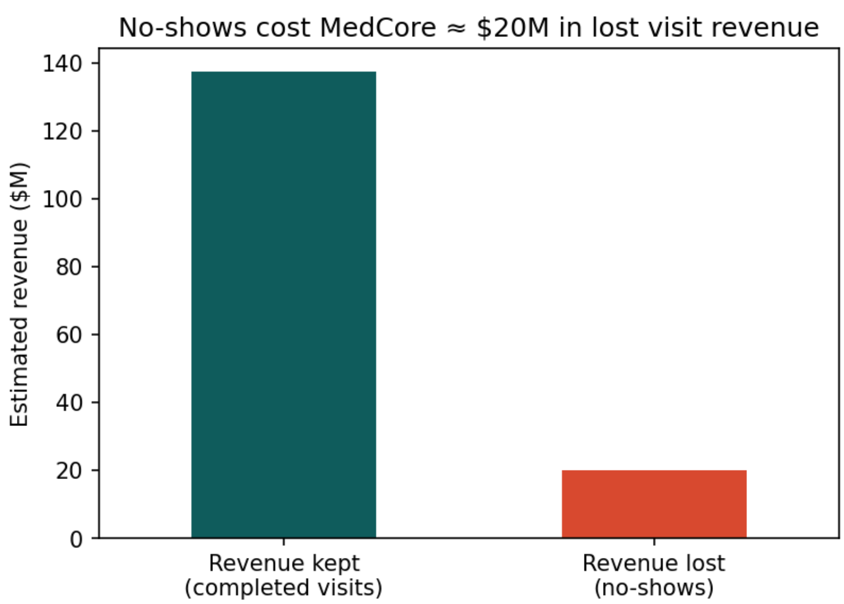

## 8. Booking Delay by Rating

The same pattern appears when patient satisfaction is measured directly by star rating rather than the At-Risk classification. Patients giving a 1-star rating waited an average of 37.2 days between scheduling and their appointment, compared with 15.9 days for patients giving a 5-star rating—more than 2.3 times longer. This reinforces the finding that longer booking delays are strongly associated with lower patient satisfaction.




## 9. The Financial Cost of No-Shows

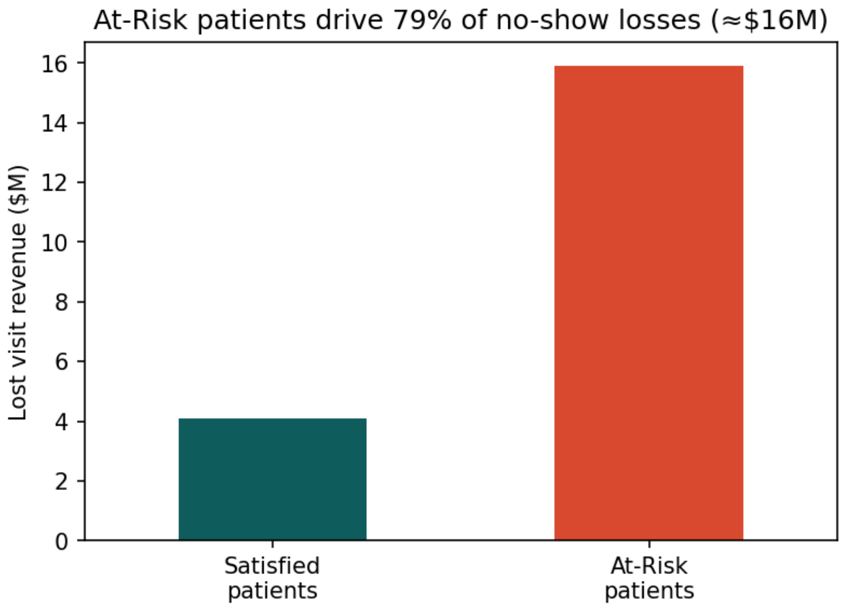
Using MedCore's billing data, the average completed visit generates approximately $617 in claims revenue. Across the three-year study period, 32,411 no-shows represent an estimated $20.0 million in lost visit revenue, or approximately $6.7 million per year.




This estimate reflects the total cost of all no-shows, regardless of patient satisfaction. The following section isolates the excess no-shows associated with patient dissatisfaction to estimate the portion of lost revenue specifically attributable to the At-Risk population.

## 10. The Cost of At-Risk No-Shows, Specifically

At-Risk patients account for 79.5% of all no-shows in the dataset (25,755 of 32,411), representing an estimated $15.9 million in lost visit revenue over the three-year study period.




However, some level of no-shows is expected regardless of patient satisfaction. Satisfied patients no-show at a baseline rate of 3.8%, indicating that not all missed appointments are attributable to dissatisfaction.

To estimate the incremental cost of dissatisfaction, we calculate the excess no-show rate among At-Risk patients above this baseline (23.2% − 3.8% = 19.4 percentage points) and apply it to the At-Risk population. This yields an estimated 21,565 excess no-shows, equivalent to approximately $13.3 million in lost visit revenue over three years, or $4.4 million annually.

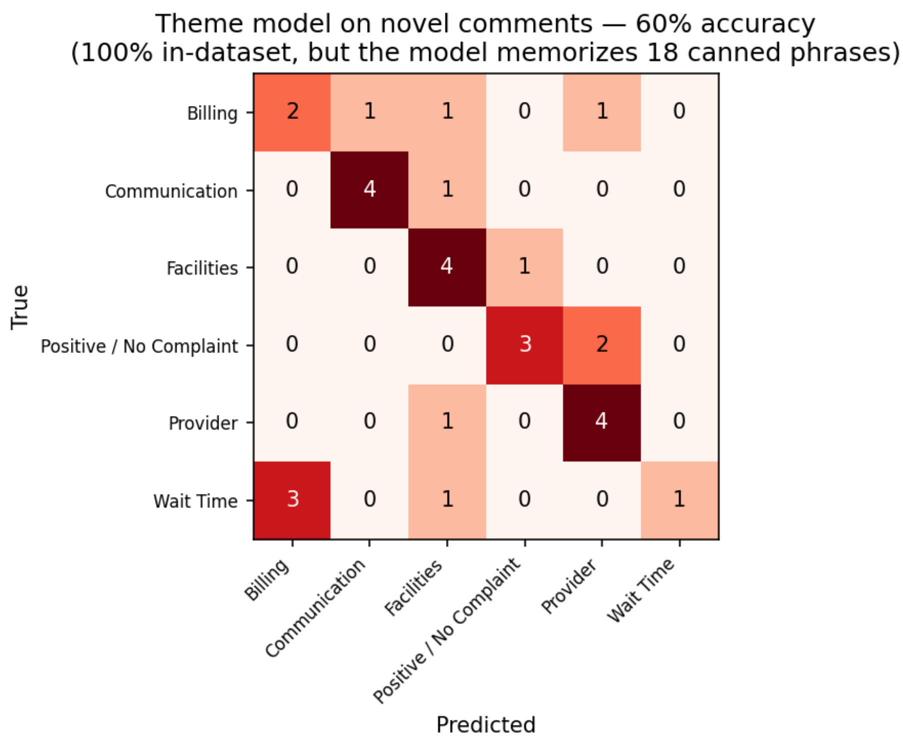
This estimate provides a more conservative and defensible business case because it isolates the marginal revenue loss associated with patient dissatisfaction rather than attributing all no-shows to the patient experience.

## 11. Appendix — Complaint Theme Classification Model

Complaint comments were classified into themes using a TF-IDF + Multinomial Naive Bayes text classifier trained on the dataset's existing complaint_category labels. Because the raw dataset contained only 18 unique canned complaint phrases, there was insufficient vocabulary to train a model capable of generalizing to new comments. To address this limitation, the training set was supplemented with 17 analyst-written example sentences covering the same five complaint categories to provide a broader range of wording.

### Two-tier evaluation

- In-dataset accuracy: 100%. The model correctly classified all 18 original complaint phrases present in the dataset. However, because these same phrases were used during training, this result reflects memorization rather than generalization and should not be interpreted as real-world performance.
- Held-out accuracy on novel comments: 60% (macro F1 = 0.58). To evaluate generalization, the model was tested on 30 previously unseen complaint comments written in new language. Accuracy fell to 60%, providing a more realistic estimate of performance on genuinely new patient feedback.



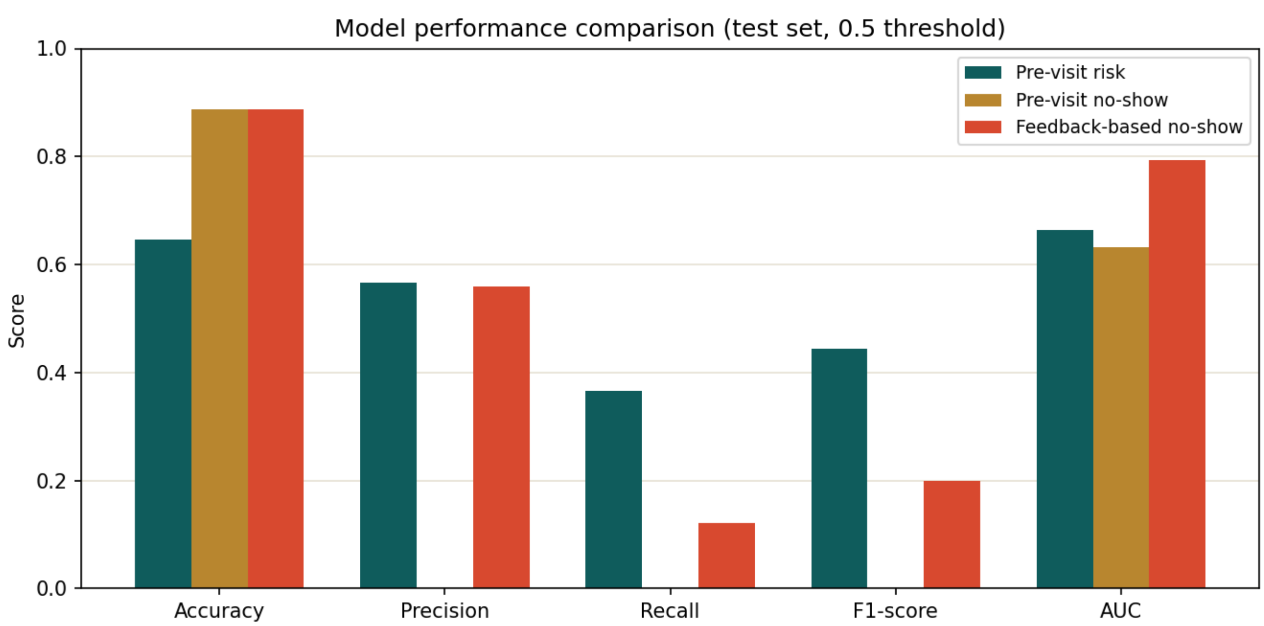

Performance varied considerably across complaint themes. Communication and Provider complaints generalized best, each achieving 80% recall on unseen comments. In contrast, Wait Time was the weakest-performing category, with only 20% recall, meaning four out of five new Wait Time complaints were classified incorrectly. Because Wait Time is also the highest-volume complaint category in the full dataset, this represents the model's most important limitation.

As a result, the classifier is appropriate for descriptive labeling of the existing 289,889 survey responses, where the vast majority of comments reuse the same limited set of 18 canned phrases. However, it is not suitable for production use on newly submitted free-text comments, particularly for Wait Time complaints. A substantially larger corpus of real, manually labeled patient comments—or a modern embedding- or LLM-based classifier—would be required before deploying automated classification for incoming patient feedback.

## 12. Predictive Models — Scoring Risk Before It Happens

Beyond labeling feedback that has already come in, we tested whether patient and visit data can predict problems in advance. Three logistic regression models were built, each targeting a different moment and a different question. All three are the same type of model: logistic regression, a supervised binary classification algorithm. It learns a weight for each input feature and outputs a probability between 0 and 1 for a yes/no outcome, which is then cut at a decision threshold (0.5 by default) to make the final prediction. What distinguishes the three models is therefore not the algorithm but the prediction task, each targets a different outcome, uses a different set of allowed features, and runs at a different point in the patient journey:

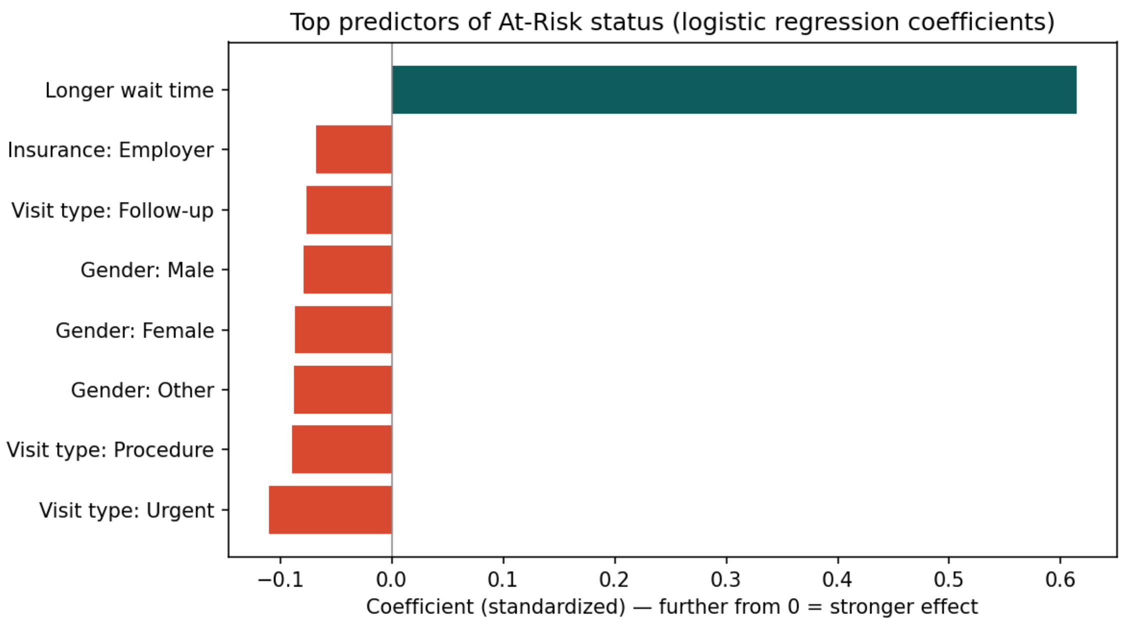
- Model 1 — Pre-visit risk: given only information known at booking (department, specialty, visit type, insurance, region, gender, chronic conditions, and department-level operating profile), predict whether the visit will end in an At-Risk rating. Rating, comment, and complaint_category are deliberately excluded as inputs, since At-Risk is defined directly from rating — including it would make the model circular.
- Model 2 — Pre-visit no-show: using the same booking-time information, predict whether the patient will no-show this appointment.
- Model 3 — Feedback-based no-show: once a patient has left a rating for a completed visit, predict whether they will no-show their next appointment, using that rating plus wait time and visit type.




| Model | Accuracy | Precision | Recall | F1 | AUC |
|---|---|---|---|---|---|
| 1 — Pre-visit risk | 0.647 | 0.567 | 0.366 | 0.445 | 0.665 |
| 2 — Pre-visit no-show | 0.887 | 0.000 | 0.000 | 0.000 | 0.633 |
| 3 — Feedback-based no-show | 0.888 | 0.558 | 0.121 | 0.200 | 0.793 |

## Model 1 — Pre-visit risk

This model achieved moderate performance (accuracy 0.647, AUC 0.665), meaningfully better than chance, but limited. The single strongest predictor was wait time, consistent with the booking-delay finding in Section 6. Adding every other available feature — department, specialty, insurance, region, gender, chronic conditions, department operating profile — improved AUC only marginally over a wait-time-only baseline (0.665 vs. 0.664). In practical terms: wait time is carrying almost all of this model's predictive power on its own.




## Model 2 — Pre-visit no-show

This model's 0.887 accuracy is an illusion: it predicted attendance for nearly every patient, earning a high score from the majority who show up while catching none of the no-shows, hence, zero precision, recall, and F1. Its AUC of 0.633 shows there is some real signal in the features, but the default 0.5 threshold is the wrong cutoff for this kind of imbalance.

> ***Accuracy is misleading here. A model that never flags a no-show can still look 89% accurate purely because most patients show up. Precision, recall, and F1 — not accuracy — are the metrics that matter for a rare-event prediction like this one.***

## Model 3 — Feedback-based no-show

This model showed the strongest discrimination of the three (AUC 0.793), indicating that a patient's own rating carries real signal about whether they'll return. But at the standard 0.5 threshold, recall was only 0.121 — meaning roughly 7 out of 8 patients who actually went on to no-show were not flagged. The model can tell at-risk patients apart from satisfied ones reasonably well in aggregate; it is simply set to a threshold that is too conservative to catch most of them individually.

> ***Overall: patient feedback (Model 3) carries the strongest predictive signal for future no-shows, and wait time is the dominant lever in pre-visit risk (Model 1). None of the three models is ready to run unattended — Models 2 and 3 in particular need class-imbalance handling (e.g. class weighting or resampling) and a***

> ***lower, tuned decision threshold before they'd be useful for flagging individual patients rather than describing patterns in aggregate.***

## Business Impact

Reducing no-shows caused by dissatisfaction could recover an estimated $4.4M per year. Beyond the direct revenue impact, giving managers visibility into problems as they happen — rather than waiting for periodic reports — means issues can be addressed while they're still small. The pre-visit risk model also makes it possible to identify patients who may be at higher risk of dissatisfaction before their appointment, allowing staff to prioritize scheduling improvements or proactive outreach. Importantly, none of this requires additional headcount, the data already exists within MedCore's systems, it simply needs to be put to use.

## Implementation and Future Outlook

The AI model currently needs more real patient comments, particularly for Wait Time complaints, before it can reliably classify new feedback on its own. For now, the model is best used to analyze feedback that has already come in, rather than to auto-classify new comments as they arrive. Because the model isn't fully trained yet, someone will need to be assigned to monitor the dashboard and respond to feedback directly during this period. Rather than rolling this out organization-wide immediately, we recommend starting with a pilot in 2–3 departments and expanding from there once the approach is validated.

As more patient comments are collected over time, the underlying classifier can be upgraded to a stronger model, such as one based on LLMs or embeddings, improving its ability to handle new, unseen phrasing. The system could also be extended to send real-time alerts when complaints spike, allowing managers to respond immediately rather than after the fact. There's an opportunity to share best practices between high- and low-performing departments, since the data shows some departments are already achieving meaningfully lower At-Risk rates under the same operating conditions. Longer term, the goal is to move beyond predicting risk and toward recommending specific scheduling improvements before patients become dissatisfied in the first place.

## Limitations

- Findings are based on observational data and demonstrate associations rather than causation.
- Complaint themes were derived from a limited set of standardized comments, restricting the classifier's ability to generalize.
- Predictive models have not yet been optimized for class imbalance and therefore should not be used operationally without further development.

## Summary of Recommendations

- Treat booking delay as a primary lever: reducing scheduling lag is likely to reduce At-Risk rates given the monotonic relationship observed (Sections 6 & 8).

- Investigate the highest-risk departments (D021, D027, D037, D025, D046) for specific operational practices driving rates 1.5–2x above lower-risk departments performing the same function.
- Prioritize Billing, Provider, and Facilities complaints for rapid response given their 100% conversion to At-Risk, even though they are lower volume than Wait Time or Communication.
- Use the $4.4M/year excess no-show cost as the basis for any ROI case for patient-experience investment, since it isolates the true marginal cost of dissatisfaction.
- Do not rely on the current theme-classification model for new, unreviewed patient comments until it is retrained on a larger, real labeled dataset — particularly for Wait Time comments.
- Before deploying either no-show model, address class imbalance and tune the decision threshold below 0.5 — as built, Model 2 misses every no-show and Model 3 catches only about 1 in 8.